Session 20 · Tuesday 3 November 2026 · A home guide. The first one is worked for you; by the fourth the scaffolding is gone. Answers are released after the session.

The network solved by hand, four times

Nine reactions, six balanced metabolites. Everything below is done with the same six lines, and a solver is never needed:

REACTION		CARBON
v_{uptake}	$\longrightarrow$ G6P, capped at 10	source
v_{pgi}	G6P $\longrightarrow$ F6P	6 → 6
v_{zwf}	G6P $\longrightarrow$ P5P + CO_2	6 → 6
v_{tpi}	F6P $\longrightarrow$ 2 T3P	6 → 6
v_{lower}	T3P $\longrightarrow$ PYR + ATP	3 → 3
v_{ackA}	PYR $\longrightarrow$ Ace + CO_2 + ATP	3 → 3
v_{tca}	PYR $\longrightarrow$ 3 CO_2 + 3 ATP	3 → 3
v_{biomass}	2 T3P + P5P + 4 ATP $\longrightarrow$	sink
v_{atpm}	ATP $\longrightarrow$	sink

The only tool is $S\mathbf{v} = 0$: for every balanced metabolite, what is made equals what is used. Write b for v_{biomass} throughout.

At each transition the sheet asks *why does that step follow?* Answer in writing before you go on.

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1 · Fully worked — the wild type

P5P. One producer, v_{zwf}. One consumer, biomass, which takes one P5P per unit. So $v_{\text{zwf}} = b$.

G6P. Made at 10, used by two reactions: $10 = v_{\text{pgi}} + v_{\text{zwf}}$, so $v_{\text{pgi}} = 10 - b$.

F6P. One producer, one consumer, so $v_{\text{tpi}} = v_{\text{pgi}} = 10 - b$.

T3P. Made at $2v_{\text{tpi}}$, used by biomass (2 per unit) and by v_{lower}:

$$2(10 - b) = 2b + v_{\text{lower}} \quad \Longrightarrow \quad v_{\text{lower}} = 20 - 4b$$

ATP. Every pyruvate goes round the TCA, which pays best. Each T3P through v_{lower} yields one ATP and then three more:

$$4(20 - 4b) \;\geq\; 4b \quad \Longrightarrow \quad 80 - 16b \geq 4b \quad \Longrightarrow \quad b \leq 4$$

So $b = 4$, and the rest falls out: $v_{\text{zwf}} = 4$, $v_{\text{pgi}} = v_{\text{tpi}} = 6$, $v_{\text{lower}} = 4$, $v_{\text{tca}} = 4$, $v_{\text{ackA}} = 0$, $v_{\text{atpm}} = 0$.

Check it. ATP made $= 4 + 3(4) = 16$; ATP used $= 4b = 16$. Nothing spare, nothing short.

▶ **Why does that step follow?** The branch at pyruvate went *all* to the TCA and none to acetate. Nothing in $S\mathbf{v} = 0$ says it must. What made the split all-or-nothing?

2 · Last step blank — knock out the TCA

Set $v_{\text{tca}} = 0$. Every line above still holds down to the ATP balance, which is the only one that mentioned the TCA. Now all pyruvate goes to acetate, which pays **one** ATP where the TCA paid three — so each T3P is worth two ATP rather than four:

$$_____ \cdot (20 - 4b) \;\geq\; 4b$$

▶ $b \leq$ _____, which is ____**% of the wild type.**

▶ **Why does that step follow?** The flux map is not the wild-type map scaled down — v_{ackA} goes from carrying nothing to carrying everything. Say in one sentence why a *linear* objective moves the answer to a different corner rather than shrinking it in place.

3 · More blank — counting what is left over, and what kills it

(a) Degrees of freedom. S has one row per balanced metabolite and one column per reaction.

Rows: _____ Columns: _____ rank S = _____

$$\dim \text{null}(S) \;=\; n_{\text{rxn}} - \text{rank}(S) \;=\; _____$$

▶ **Why does that step follow?** You found three directions the stoichiometry cannot choose between. Session 3 wrote a null-space dimension too, and got a different formula. In one sentence: what was *its* n counting, and why is it not this one?

(b) Lethality by inspection. Of the seven real reactions, four deletions drive b to exactly zero. Name them, and for each name *the one thing biomass needs that nothing else can make*.

DELETED	$b =$	BECAUSE NOTHING ELSE MAKES

▶ **Why does that step follow?** Two of the remaining three cost nothing at all. Which two, and what do they have in common at the optimum?

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4 · Entirely yours — the answer stops being unique

Add one reaction, v_{zwf2}, with **exactly** the stoichiometry of v_{zwf}: G6P ⟶ P5P + CO_2. Nothing else changes.

(a) What is the optimal b now? Justify it in one line, without solving anything.

(b) The solver returns $v_{\mathrm{zwf}} = 4$, $v_{\mathrm{zwf2}} = 0$. Write down a *different* flux vector that is just as optimal. Then a third.

(c) How many optimal flux vectors are there? Describe the set in one sentence, and say whether every member of it is a vertex of the feasible region.

(d) Flux variability analysis re-solves for the smallest and largest value each flux can take while b stays optimal. What does it return for v_{zwf}, and what for v_{pgi}? Why are the two answers different in kind?

(e) In §3(b) you found v_{zwf} and b locked to each other. Is that still true here? What broke it, and what does that tell you about where coupling comes from?

Before Thursday

Mishra *et al.*, *Nat Biotechnol* 32:1268 (2014) — a load driver. A circuit slows down because something downstream is *reading* it. Read Figure 1 and the rise times in the Online Methods.

Come able to say what the "load" physically is.